Sport Type Classification of Mobile Videos

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***Abstract*—The recent proliferation of mobile video content has emphasized the need for applications such as automatic organization and automatic editing of videos. These applications could greatly benefit from domain knowledge about the content. However, extracting semantic information from mobile videos is a challenging task, due to their unconstrained nature. We extract domain knowledge about sport events recorded by multiple users, by classifying the sport type into soccer, American football, basketball, tennis, ice-hockey, or volleyball. We adopt a multi-user and multimodal approach, where each user simultaneously captures audio-visual content and auxiliary sensor data (from magnetometers and accelerometers). Firstly, each modality is separately analyzed; then, analysis results are fused for obtaining the sport type. The auxiliary sensor data is used for extracting more discriminative spatio-temporal visual features and efficient camera motion features. The contribution of each modality to the fusion process is adapted according to the quality of the input data. We performed extensive experiments on data collected at public sport events, showing the merits of using different combinations of modalities and fusion methods. The results indicate that analyzing multimodal and multi-user data, coupled with adaptive fusion, improves classification accuracies in most tested cases, up to 95.45%.**

***Index Terms*—Fusion, mobile, sport, video.**

I. Introduction

THANKS to the ubiquity of camera-enabled mobile devices, such as mobile phones, it has become common for users to record videos of events they find interesting at any time and virtually anywhere. We refer to these recordings as *user generated videos* to distinguish them from professional audio-visual content, i.e., from content recorded by professional camera setups such as for TV content production. In particular, one example of user generated videos is represented by those videos recorded using mobile phones, and which are also referred to as *mobile videos*. Moreover, there are situations in which multiple people record the same scenery during the same time span, such as during sport events. By considering such a scenario, in this paper we propose an approach for automatically classifying mobile videos according to the sport type (e.g., soccer, tennis, etc.).

Videos recorded by multiple users and capturing the same scenery could be automatically combined for producing multi-camera video summaries. This would require analyzing the content in order to understand the semantics therein, and detect the key highlights. Generic content-based analysis of sport video is challenging because of the diversity of sport rules [1]. Thus, it is convenient to first classify the videos according to the sport type, and then the subsequent analysis steps can exploit domain knowledge, for example by selecting combination rules which are specific to the identified sport type. Moreover, sport type classification is useful also for other purposes such as video search applications and automatic organization of personal media collections.

Our system categorizes sport events according to the following sport types: *soccer*, *American football*, *basketball*, *tennis*, *ice-hockey* and *volleyball*. To achieve effective classification, we analyze multiple data modalities which are captured by multiple users. In particular, we propose to analyze both spatial and spatio-temporal *visual* information, *audio* data and data captured by *auxiliary sensors* embedded in mobile devices. The considered auxiliary sensors are magnetometer and accelerometer.

This paper is an extended version of our previous work [2], in which our main contributions were: extraction of novel camera motion features from auxiliary sensors and their utilization for sport type classification, and dynamic adaptation of the fusion of different modalities based on their quality for each video. The additional contributions of this paper are as follows: (i) Auxiliary-sensor-aided extraction of spatio-temporal visual features that are *less affected by camera motion*. (ii) Use of the extracted *spatio-temporal visual features* for inferring the sport type. (iii) Use of separate *bag-of-visual-words* models for representing the spatial and temporal aspects of spatio-temporal features, which allows for a separate optimization of vocabulary sizes. (iv) Feature weighting by using modality qualities, for the purpose of dynamically adapting *early* and *intermediate* fusion approaches based on the input data. (v) Use of *cross-modal qualities*, for dynamically adapting the fusion process, i.e., the quality of the auxiliary sensor modality is used as a measure for the quality of the visual spatio-temporal features. (vi) A novel late fusion method, namely *2D majority voting*. (vii) An extensive comparison among different schemes of multi-class Support Vector Machines (SVM) in the fusion process, including one version which fuses different modalities separately for each class. (viii) A comprehensive analysis and comparison of several scenarios, including fusing all possible modality-combinations using all the fusion methods considered

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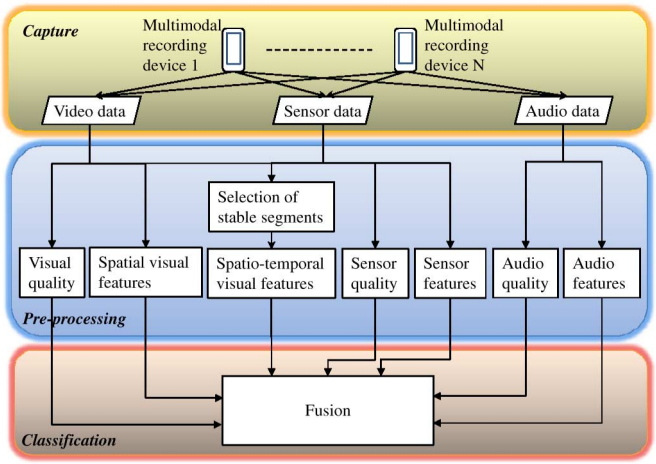


Fig. 1. Overview of the proposed approach.

in this paper, impact of viewing angle on the classification accuracies, and per-user classification accuracy.

Our approach is not limited to mobile videos but it can be directly applied to videos captured by other types of recording devices embedding the auxiliary sensors considered in this work, such as some models of hand-held cameras.

An overview of the workflow is given in Fig. 1. In the *capture* step, we consider multimodal data captured by multiple mobile phone users at a sport event. In particular, multiple videos (and associated audio and sensor data) belong to the same event. The *pre-processing* step consists of the feature extraction and modality quality estimation. Regarding the *classification* step, we consider three main approaches: *early fusion*, *intermediate fusion*, and *late fusion* of modalities.

The paper is organized as follows: Section II introduces the related prior art, while Section III describes the data capture. Feature extraction and modality quality estimation are discussed in Section IV, and the classification schemes are described in Section V. Section VI contains the experimental analysis and discusses the achieved results. Finally Section VII concludes the paper.

II. Prior Art

Content-based analysis of videos for the purpose of classification has been studied extensively–a survey on this topic is presented by Brezeale *et al.* in [3]. Videos can be classified for example according to genres, such as *movie*, *commercials*, *news*, *sport*, *music*, and sub-genres, as pointed out for example by Yuan *et al.* [4]. For example, movie genre can be sub-divided into *action*, *comedy*, etc., and sport genre into *soccer*, *basketball*, etc. We focus on sport sub-genres, and, for the sake of simplicity, we refer to them just as *sport types*. Examples of existing works on sport type classification are described in [1], [4], and [5]. However, these works consider only video content recorded by professional TV producers. In contrast, we consider *user generated content* which is highly *unconstrained* and mostly *unedited* [6]. In *unconstrained* videos, cameras are often moved in uncontrolled ways by non-professional users, the viewpoint of the recorded scenery is rarely optimal, and the audio-track is likely to be corrupted by background noise such as cheering. We propose to cope with these issues by extracting motion features taking into account camera motion and shake, by considering data captured from multiple viewpoints, and by utilizing information about the quality of the input data. Furthermore, the considered content consists of raw *unedited* video sequences. This means that user generated videos cannot be classified by exploiting the presence of editing patterns such as replays, scoreboards and specific shot-types. Thus, we rely only on features and semantics present in the raw data.

To the best of our knowledge, the only work on genre classification of user generated content is the one of Sugano *et al.* [7]. This work considers features such as camera motion and subject motion (from motion vectors), audio class, audio volume, luminance, color and flashlight, for discriminating among the genres *travel*, *sports*, *family & pets*, *event* and *entertainment*. However, the authors do not further classify the sub-genres. Other authors have proposed to perform content-based camera motion estimation for the task of classifying the sport type but for professionally produced content [1], [4], [5]. Unlike these works, we model the camera motion by analyzing auxiliary sensor data, thus completely avoiding the high computational cost required by video content based global motion estimation. Furthermore, unlike content-based approaches, the performance of our approach is not impaired by the presence of moving objects in the recorded content.

Recently there has been a growing interest from the research community towards *local spatio-temporal visual features*, such as the ones described in [8], due to their accuracy in describing content motion. They have been successfully applied to the task of action recognition, such as in [9]. However, spatio-temporal interest points have not been used for sport type categorization, even though the movements of players and the whole game-flow are very different among sports, and thus can represent a discriminative semantic cue. Nevertheless, spatio-temporal features can be corrupted by camera motion. In order to overcome this problem, in [10] images are segmented into dominant planes which are then used for correcting the local motion vectors. Instead, in our work we propose an efficient way to limit the influence of camera motion on spatio-temporal features, by relying on auxiliary sensor data instead of visual content.

In classification tasks, the two most common approaches for combining the predictions from multiple component information sources (e.g., features or modalities) into one improved overall decision are so called *early fusion* and *late fusion* [11]. They are also known as feature level fusion and decision level fusion, respectively. These approaches aim at performing the fusion at the pre-classification and post-classification stages, respectively. Generally, early fusion requires training a reduced number of classifiers, eventually only one. However, it can suffer from restrictions introduced by the different data representations of the component information sources. Late fusion, on the other hand, usually has more consistent representations between the components, such as likelihood scores for each class, all class labels ranked according to likelihood, or the single most likely class label. Yet, it needs a separate classifier training phase for each component. In our work we evaluate the performance of both early and late fusion approaches. In [12] early fusion is performed for discriminating sport types, by concatenating histograms of local visual features and geometry

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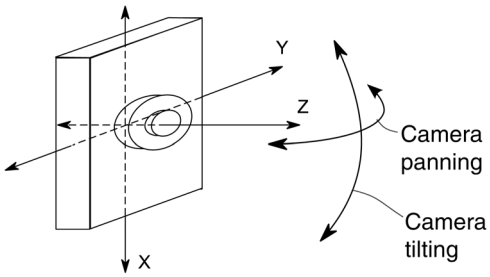


Fig. 2. Camera movements and accelerometer axes.

visual phrase before an SVM. In [13], the authors successfully applied late fusion for discriminating among six TV broadcast genres and sub-genres: cartoon, commercial, news, cricket, football, and tennis. Their approach consisted of weighting the results from the individual classifiers according to their performance.

III. Data Capture

In this work we consider smartphones which, in addition to cameras and microphones, embed the following auxiliary sensors: *magnetometers* and *accelerometers*. Whenever the meanings are not ambiguous, we will refer to the auxiliary sensors and to the whole recording device just as sensors and camera, respectively. We perform multimodal data recording by simultaneously capturing video data, audio data and data output by auxiliary sensors, which are considered in this paper as three separate modalities. The auxiliary sensor data describe the orientation and motion of the recording device and, by using features extracted from such data, we build models of camera motion for each sport type.

Our test devices are equipped with a tri-axial magnetometer which is sometimes known also as "electronic compass". Such a sensor provides the horizontal orientation at a certain sampling rate and in the form of an angular displacement from the magnetic north, measured in degrees. We use tri-axial accelerometers for capturing the acceleration of the recording device across three mutually perpendicular axes. In the absence of other acceleration, the accelerometer captures a static acceleration of approximately 9.81 m/s^2 at sea level in the direction of Earth's center of mass. We exploit this static acceleration for detecting the camera orientation with respect to the horizontal plane, i.e., the plane which is perpendicular to the gravitation force.

If the magnetometer and accelerometer sampling rates are sufficiently high, during video recording it is possible to capture sensor data densely enough to provide information about *panning* and *tilting* movements, i.e., horizontal-rotational and vertical camera movements, respectively. Fig. 2 shows the three accelerometer measurement axes and the considered camera movements.

IV. Feature Extraction and Modality Quality

From visual, audio and auxiliary sensor data modalities we extract discriminative features for the sport type classification problem. In particular, from visual data we extract both spatial features from each frame and spatio-temporal features from groups of adjacent frames. We consider these two types of visual features as two separate modalities because they are treated separately in the same way as audio and auxiliary sensor modalities.

A. Spatial Visual Features

Spatial visual features are extracted from representative frames of each input video sequence. In this work we consider the following MPEG-7 visual descriptors [14]: *dominant color*, *color layout*, *color structure*, *scalable color*, *edge histogram*, and *homogeneous texture*. By concatenating these descriptors we obtain a 365-dimensional vector. MPEG-7 features are generic features which do not exploit existing structure or editing patterns in the video footage, and thus they are suitable for user generated content. Color structure, color layout, edge histogram and homogeneous texture were already used in [15] for the purpose of sports video categorization. Additional details on our spatial visual feature extraction can be found in [2].

B. Sensor-Enhanced Extraction of Spatio-Temporal Visual Features

Different sport types are characterized by different motion patterns of the players and by different game-flows. For example, soccer players are almost constantly running and they hit the ball by using mainly their legs and feet; instead, volleyball players mainly jump (vertical motion), dive, hit the ball using their hands, and run only rarely and for short distances. We exploit these differences by extracting Spatio-Temporal Interest Points (STIP), proposed by Laptev *et al.* in [8], which describe the content motion. They are detected by extending the Harris corner detector from two spatial dimensions to consider also the time dimension. An interest point is detected when the image values have significant local variations in both space and time. Hence, STIP features capture both the appearance and motion information. For each image point a spatio-temporal second-moment matrix is computed as follows:

$$\mu\left(\cdot;\sigma,\tau\right)=g\left(\cdot;s\sigma,s\tau\right)*\left(\nabla L\left(\cdot;\sigma,\tau\right)\left(\nabla L\left(\cdot;\sigma,\tau\right)\right)^{T}\right), \quad (1)$$

where σ and τ are independent spatial and temporal scales, respectively, g is a separable Gaussian smoothing function and ∇L are the space-time gradients. A space-time interest point is given by the local maxima of $H=\det(\mu)-k\,trace(\mu)$, where *det* is the determinant and *trace* is the sum of the elements on the main diagonal of the matrix μ. We used the default parameter values, which are $k=0.0005$, $\sigma^2=4,8,16,32,64,128$, $\tau^2=2,4$. Given the large amount of data considered in our work, the sparsity of STIP features is a desirable property for keeping the computational cost low in the subsequent clustering step, which is described below.

Our content is heavily affected by both intentional and unintentional camera motion, where the unintentional motion is mainly due to shake or to accidental situations. As a consequence of such motion, the extracted STIP features would describe not only the motion of objects/people, but also the effect of camera motion on the recorded content. We propose to overcome this limitation by having a combined analysis of sensor data and visual data. As shown in Fig. 3, by analyzing sensor data, we identify stable video segments, i.e., segments not affected by camera panning, tilting or shake, which are detected

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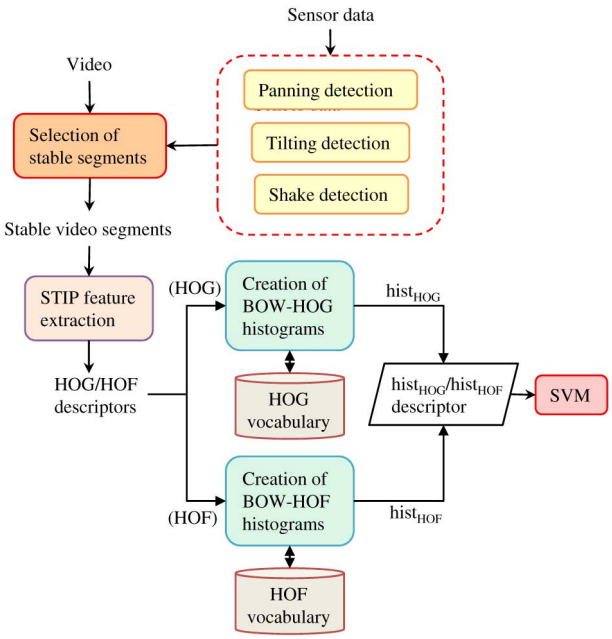


Fig. 3. The proposed sensor-aided extraction of spatio-temporal visual features. Features are extracted only from stable segments, identified by analyzing accelerometer and magnetometer data. Spatial and temporal information are represented by using two separate BOW models.

as described in [16]; then, we extract STIP features only from these segments. In order to perform a fair comparison of the performances obtained by extracting features only from stable segments and by extracting them regardless of the camera steadiness, the length of analyzed data should be the same. Thus, for the latter case we consider segments randomly distributed within each video and whose lengths sum up to the total length of stable segments.

Because of the position and angle that mobile phone cameras usually have with respect to a sport field which is being recorded, the videos often include content which does not belong to the field or to the players. For example, spectators who are located on the opposite side of the field often appear at the top of the video frames; also, those spectators on the same side and in front of the recording camera often appear at the bottom of the video frames. Hence, since frame borders would often bring only noise and no useful information, we discard interest points extracted from borders of width equal to 1/6 of either the frame width or the frame height.

The detected STIP features are commonly described using Histogram of Oriented Gradients - Histogram of Optical Flow (HOG-HOF) descriptors. In order to obtain the same number of descriptors per video we use the Bag-of-Words (BOW) model [17], which is adapted for obtaining histograms of *spatio-temporal visual words* of the BOW vocabulary. In particular, as shown in Fig. 3, we use two separate BOW setups for the HOG and the HOF portions of the descriptors, as this strategy proved to give higher classification accuracies–we will show it in the experimental section. To this end, two vocabularies need to be generated. One advantage of this approach is that separate and thus *more specific BOW words* can be used for the HOG and for the HOF descriptors; another advantage is that *different vocabulary sizes* can be used for HOG and HOF, which can be optimized separately. Clustering is performed by using the k-means algorithm. Each space-time point is assigned to its closest vocabulary word using the Euclidean distance. Optimization of the vocabulary size is a crucial step in the BOW approach. In fact, as stated in [17], the vocabulary should be large enough to distinguish relevant changes in image/video parts, hence allowing for good discriminability. At the same time, it should be not so large as to distinguish irrelevant variations such as noise, which would result in low generalizability. Since we are not interested in obtaining optimal clusters, e.g., based on within/between cluster scatter, but in high classification accuracy, we evaluated the classification performance for every possible combination pair of sizes k for the two HOG and HOF vocabularies, where $k \in \{500, 1000, 1500, 2000, 2500, 3000\}$. As we show in the experimental analysis section, the best performance was achieved by the combination pair (2500, 2000); therefore we used this combination for the subsequent steps, i.e., in the modality fusion.

The proposed sensor-aided extraction of spatio-temporal visual features is *efficient* for two reasons. On one hand, the lower influence of camera motion is obtained in a light-weight manner due to the light-weight nature of auxiliary sensor data. On the other hand, we avoid extracting features from a large portion of the whole visual data, represented by the *non-stable* segments, thus heavily decreasing the overall computational cost. Details about both of these aspects are provided in the experimental analysis section.

C. Audio Features

From the audio-track contained in each recorded video, we extract Mel-Frequency Cepstral Coefficient (MFCC) features. MFCC are widely used in audio analysis for many purposes such as speech recognition, audio and music classification and retrieval [18], [19]. They have been used successfully for example in audio-based environment recognition, which is closely related to the recognition task considered in this paper [20]. The MFCCs are calculated from monophonic audio data recorded with a 44.1 KHz sampling rate. Feature vectors with 12 MFCC coefficients each are calculated from 40 ms frames with no overlap between frames.

D. Auxiliary Sensor Features

As already observed for the spatio-temporal visual features, the game-flow and the players' movements are different for different sports. This does not affect only the visual content but also how the cameras are moved by their users. For example, in an ice-hockey game the players move very fast, and therefore users need to follow the action by quickly panning their cameras. Instead, in volleyball the action stays for a certain amount of time (up to 4-5 seconds) on one side of the field before passing to the other side where it stays for another similar amount of time; hence, some of the cameras are moved accordingly to this pattern.

There are mainly two factors which "force" the users to use camera motion. First, the limited *angle of view* of mobile phone cameras does not allow following the main action without moving the camera at all. In our experiments, the average distance between cameras and the soccer field was about 15

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meters, and the longer side of the field was about 100 meters long. Thus, covering the whole longer side of the field without moving the camera would require a horizontal angle of view of 146.6 degrees. Instead, the angle of view of our test phones was only 39.7 degrees, which allows for covering a maximum horizontal length of about 10.8 meters; thus camera motion is needed for covering the whole field. The second factor is that users often adjust the orientation of their cameras in order to place the main action roughly at the center of the camera view-finder. Based on these observations, camera motion is a potentially discriminative cue for sport types.

We model the camera motion by analyzing data captured by auxiliary sensors. Data captured by the magnetometer during the recording of a video is analyzed for extracting the following horizontal camera motion features: *panning rate*, computed as the frequency of panning movements within one video; average, median, minimum and maximum values of *panning extents*, *panning speeds* and *panning durations*, where panning extent is the angular width of a panning movement; *panning direction change rate*, computed as the rate with which two subsequent panning movements change direction, i.e., from clockwise to counter-clockwise, or vice-versa; *Discrete Cosine Transform (DCT) components*, computed from the magnetometer data stream. We extract also the *magnetometer variance* feature, computed as the variance of all the horizontal camera orientations, which can be informative of the size of the playing field and hence it can help to discriminate sports with large fields, such as soccer and ice-hockey, from sports with smaller fields, such as tennis. From accelerometer data we extract the following vertical camera motion features: *tilting rate*; average, median, minimum and maximum values of *tilting speeds* and *tilting durations*; *accelerometer DCT components*; *accelerometer variance*. All these features are concatenated into a 31-dimensional vector. More details about these auxiliary sensor features can be found in [2]. Camera panning and tilting are determined from magnetometer and accelerometer data, respectively, as described by Cricri *et al.* in [16].

E. Modality Qualities

In "clean" recording conditions, most commercially available smartphones are able to record audio and video without big issues in quality. By "clean" we mean that there are good illumination conditions, no sources of audio noise, and the cameras are static or without significant motion. However, in this paper we consider a specific recording scenario, i.e., sport events held either indoors or outdoors, in which the illumination conditions may vary considerably, there are usually many spectators nearby the recording users, and the cameras are moved in unconstrained way. All these aspects may cause artifacts in the captured data, which may corrupt the extracted features and thus the subsequent sport type classification process.

We propose to adapt the contribution of each modality to the final decision based on the quality of that modality in the given input data, where the quality expresses the *reliability* of the modality. For example, the audio extracted from a certain video should contribute less if it is affected by audio noise. To this end, quality estimates of each modality need to be computed. In particular, we are interested in measuring the *quality of only the considered features* (e.g., color). Furthermore, audio and static visual features are extracted from frames which are uniformly spaced over the whole length of each video. Therefore, the classification of each video is not significantly corrupted by artifacts affecting only a single or few frames or, in the case of visual data, by artifacts affecting only limited spatial regions of a frame. For example, motion blur caused by fast ball movements would affect visual data only in a limited temporal and spatial portion. Based on these observations, we need to consider only those artifacts which *affect a relevant portion* of the data associated to each video and, thus, a relevant portion of the extracted features.



Fig. 4. A dark frame captured by a mobile phone at one of the American football events in our datasets. In this case, the darkness is caused by back-light from the Sun.

We compute an estimate of the quality of each modality for each input video. The qualities of the sensors, audio and spatial visual modalities are derived from *sensor calibration* level, *audio noisiness* measure and *visual luminance* measure, respectively. Details about these qualities are given in the following paragraphs.

The quality of the magnetometer data is approximately proportional to the calibration level of the magnetometer–see [2] for more details. Thus, we set the quality of sensor data q_s equal to this calibration level after being scaled in the range [0, 1].

Regarding the audio signal, we need to take into account the audio sources present at sport events which may cover the characteristic sounds of each sport type–one example of a characteristic sound for basketball is generated by a bouncing ball. There may be many potential sources of audio disturbance in sport events. One example for outdoor events may be wind; however, in our dataset this was not an issue, probably because the sport venues were well protected from rain and wind in order to make the watching experience of spectators more comfortable. The main source of audio disturbance was empirically found to be the cheering from spectators, who are usually in the vicinity of the recording phones. Therefore we classify segments of the audio track as *cheering* and *non-cheering*–see [2] for more details on how this classification is performed. The audio noisiness level n_a is computed as the portion of the audio track containing cheering, and can range from 0 to 1. The audio quality level is obtained as $q_a = 1 - n_a$.

Regarding the visual data, based on visual inspection we found out that artifacts such as motion blur or blocking artifacts are not present. Instead, the main visual artifact is represented by dark content. This may be caused either by poor illumination conditions, e.g., if the weather is very cloudy, or by back-light for example due to the Sun. One example of a dark frame caused by back-light at one American football game is shown in Fig. 4. As can be seen from this figure, dark content is

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likely to corrupt the color information and thus the extracted color features, but also textural information. Furthermore, in the presence of very low illumination conditions, the recorded frames may be affected by sensor noise, which can corrupt the texture. Based on these observations, we decided to estimate the quality of the spatial visual features according to the presence of dark content. To this end, several algorithms may be used; however, since we need to analyze many videos, we opted for a computationally lightweight approach. Visual data from each video is analyzed in order to extract the luminance level [21], computed as follows:

$$L = 0.2126R + 0.7152G + 0.0722B \tag{2}$$

where R, G, B are the linearized RGB components, obtained by applying the inverse of the gamma function to the gamma-compressed R'G'B' components. The final luminance level of a video is obtained by averaging all luminance levels computed from all pixels of all representative frames. We scale the luminance level to the range [0, 1] and assign it to the spatial visual quality q_v. Therefore, a low value of the scaled luminance level indicates a low quality level of the spatial visual modality. For example, the scaled luminance value of the frame shown in Fig. 4 is 0.104.

Furthermore, we introduce a *cross-modal* quality level for the extracted STIP features, q_t, which is computed as the average of the sensor and the spatial visual qualities. STIP features are extracted only from stable segments identified by auxiliary sensor analysis. Hence, if the auxiliary sensor data is not reliable, stable segments and their extracted features are considered not reliable either.

Modality qualities are used for adapting the contribution of the corresponding input modalities in the fusion step; the specific way by which they are used is indicated by the following symbols: $\mathrm{S_Q}$–qualities drive the scaling of input features; Q–qualities drive the selection of modalities; $\mathrm{W_Q}$–qualities drive the weighting of likelihoods.

V. Classification

Each event is represented by multiple videos, which are recorded by multiple users. In order to obtain an overall predicted sport-class for the recorded event, we first classify each video by fusing the information provided by the different modalities. To this end, three main approaches are considered: (i) *Early Fusion*, in which features from all modalities are first concatenated and then given to an SVM–this fusion is simply referred to as *SVM* in the paper; (ii) *Intermediate Fusion*, in which uni-modal classification results are fused separately for each class–this fusion is referred to as *SVM-FSC*, where *FSC* stands for "fusing separately for each class"; (iii) *Late Fusion*, in which likelihoods from multiple modalities and for all classes are fused using either rule-based fusion, meta-level SVM, or meta-level SVM-FSC. In particular, regarding late fusion, the considered rule-based methods are majority voting and weighted average; the meta-level SVM, also referred to as *Meta-SVM*, is applied on concatenated likelihoods from uni-modal classifiers; instead, in the meta-level SVM-FSC, also referred to as *Meta-SVM-FSC*, classification results obtained from meta-classifiers are fused separately for each class.

Then, we aggregate the fused predictions for all videos of an event into one final event-level prediction, by using majority voting.

A. Early Fusion: Fusion of Concatenated Multimodal Features

In early fusion the features from different modalities are first concatenated and then fed into a single classifier–we consider a one-versus-all multi-class SVM classifier. The concatenation requires features with a common data rate. However, the sensor and spatio-temporal visual feature extractors provide one feature vector per video, whereas both the audio and the spatial visual feature extractors provide one feature vector per audio frame and per video frame, respectively. To obtain features at a common data rate, we average the audio and spatial visual features over all frames of each video. Regarding the classifier, the optimal SVM parameters were found by grid search and we chose the linear kernel as it yielded the best classification accuracies.

In early fusion the modality qualities for the given input data are used for driving the weighting, or scaling, of features for the corresponding modalities. We refer to this method as $SVM + S_Q$. The weighting has the effect of narrowing down the range of each feature from the original range [0, 1] to a new range $[0, \mathrm{B_H}]$, where $\mathrm{B_H} \leq 1$, hence reducing the influence or contribution of that feature in discriminating classes. Feature weighting has already been used before, such as in [22], but in our case the selection of proper weights is driven by the modality qualities and is performed *locally*, i.e., based on the given input data. We weight each feature extracted from the k-th modality of the i-th video as follows:

$$\begin{aligned} & F_{ik}^{\sigma} = \sigma_{ik} F_{ik} \\ where \quad & \sigma_{ik} = \begin{cases} 1, & if\ q_{ik} \geq q_k^{thr} \\ q_{ik}/q_k^{thr}, & otherwise, \end{cases} \end{aligned} \tag{3}$$

where F_{ik} is the original feature value, F_{ik}^{σ} is the weighted feature value, σ_{ik} is the weight assigned to the original feature value, q_{ik} is the quality level for the k-th modality in the given input video, and q_k^{thr} is a predetermined quality threshold for the k-th modality. The optimal thresholds were found by grid search.

B. Intermediate Fusion

In this approach a network of binary classifiers similar to the one proposed in [23] is used, with the difference that SVMs are used as the binary classifiers instead of the evolutionary multi-layer perceptrons (MLPs). An illustration of the network is given in Fig. 5, in which 6 classes are considered. The characteristic of this network is that results from uni-modal classifiers are fused separately for each class–this is performed in the Stage 1 Fuser. The network performs classification in two stages: (i) for each class, features from each modality are fed to a separate binary SVM which performs one-versus-all classification; then, the results obtained by the binary SVMs for a certain class and different modalities are fused by using the "Stage 1 Fuser" which is an SVM; (ii) results from the first stage for different classes are fused by majority voting in the "Stage 2 Fuser". Note from Fig. 5 that the input features from different modalities are split and given to N_c Stage 1 networks, where

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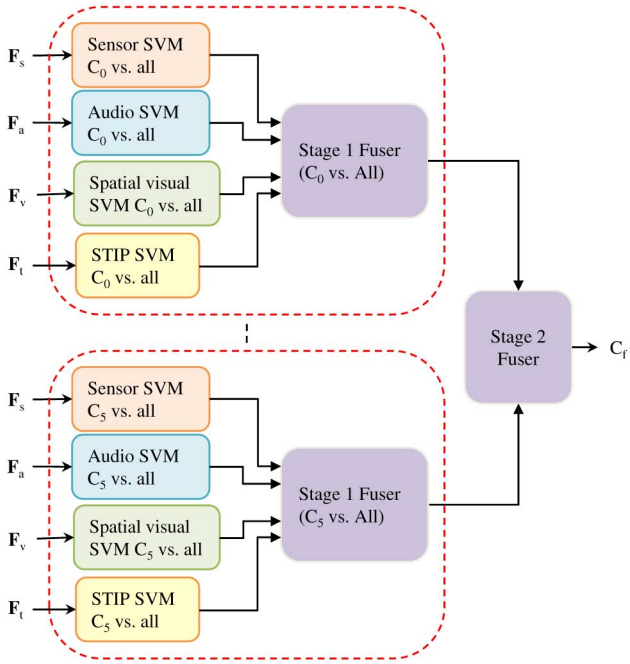


Fig. 5. *Intermediate Fusion SVM-FSC*: F_s, F_a, F_v, F_t are the features for sensor, audio, spatial visual and spatio-temporal data, respectively.

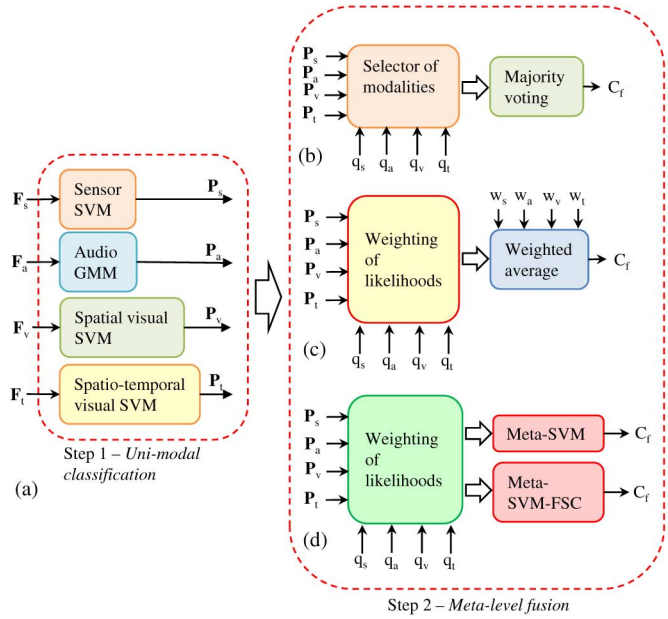


Fig. 6. Late fusion. (a) Uni-modal classification; (b) majority voting; (c) weighted average with weighted likelihoods; (d) Meta-SVM and Meta-SVM-FSC with weighted likelihoods.

N_c is the number of classes–6 in our case. A detailed description of the Stage 1 and 2 is given in [23]. All SVM parameters are optimized by grid search.

The main difference of this approach with respect to applying the meta-SVM, which is discussed in the next subsection, is that the decisions from different modalities are fused for each class separately, against all other classes. We refer to the intermediate fusion as *SVM-FSC*, where *FSC* means that fusion of modalities is done separately for each class. In *SVM-FSC*+S_Q the modality qualities are used for scaling the features, as in the early fusion.

C. Late Fusion

As illustrated in Fig. 6, late fusion consists of two steps: (i) uni-modal classification, where each modality is independently classified by considering all classes; (ii) fusion, where the likelihoods output by each uni-modal classifier are fused to obtain a single class prediction.

First Step: Classification of Uni-Modal Features: Spatial and spatio-temporal visual features are classified using two SVMs with polynomial and linear kernel, respectively. Camera motion models are built by feeding auxiliary sensor features to another SVM with Radial Basis Function (RBF) kernel. The choice of the classifier and kernel types was done through empirical evaluation. Originally, SVMs output only class predictions, not probabilities. We use a publicly available implementation of SVM, LIBSVM [24], in which probability estimates are obtained as described in [25].

The audio scenes are modeled using Gaussian mixture models (GMMs) which are a common choice when classifying audio data. Expectation-maximization (EM) training was used to obtain GMMs with 16 Gaussian densities for each sport type. The number of Gaussians per GMM was increased during training by running a few iterations of EM training, then splitting the Gaussians with the largest variance and repeating this a few times. The EM training was done using the HTK hidden Markov model toolkit [26].

Second Step: Fusion of Classifier Outputs: The class likelihoods which are output by the uni-modal classifiers are fused by considering the following fusion methods: rule-based fusion and meta-level classifier. The considered rule-based methods are majority voting and weighted average. A "meta-level" classifier receives as input the likelihoods that are output by the "base-level" classifiers–in our case these are uni-modal classifiers, as each of them analyzes features from only one data modality. We propose to integrate the modality quality levels within the fusion process as shown in Fig. 6.

In *majority voting + Q* fusion we consider the modality quality levels for selecting the modalities to be fused. If a certain modality has a low quality level with respect to a predefined modality-specific threshold, that modality will be discarded from the majority computation. If there is no majority, we assume that modalities have the following descending priority order, which is derived from the uni-modal cross-validation accuracies: (i) spatial visual, (ii) audio, (iii) spatio-temporal visual, and (iv) auxiliary sensors. For example, if for a certain input video there is no majority and the spatial visual modality has a low quality level, we use the audio prediction.

We consider also a variation of the majority voting method, that we call *2D majority voting*. This method does not provide a video-level prediction. Instead, it directly provides an event-level prediction by performing majority voting over the crisp predictions of *all modalities* and *all videos* of an event, which are collected into a two dimensional array (i.e., a matrix) for majority computation. In 2D majority voting + Q, the modality quality levels are used in the same way as in majority voting fusion.

In the *weighted average* fusion, the weight of each modality is set according to the corresponding uni-modal accuracy–this accuracy was obtained by performing cross-validation on each

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uni-modal classifier, and by classifying single videos instead of events. In the *weighted average* + W_Q fusion we use the modality qualities for *weighting the input likelihoods*: if, for a given input video, the quality of a certain modality is low with respect to a predefined threshold, the corresponding classifier likelihoods are weighted towards uniform likelihoods in order to de-emphasize this modality in the fusion. Uniform likelihoods are probabilities from a uniform distribution. More precisely, the likelihood of video v_i and class C_j, output by a uni-modal classifier for the k-th modality, is weighted in the following way:

$$P_k^w(C_j|v_i) = w_{ik}P_k(C_j|v_i) + (1 - w_{ik})P_u$$
$$\text{where} \quad w_{ik} = \begin{cases} 1, & if\ q_{ik} \geq q_k^{thr} \\ q_{ik}/q_k^{thr}, & otherwise. \end{cases} \tag{4}$$

In (4) $P_k^w(C_j|v_i)$ is the weighted likelihood, $P_k(C_j|v_i)$ is the original likelihood, $P_u = 1/N_c$ is the probability from a uniform distribution (N_c being the number of classes), w_{ik} is the weight applied to the original likelihood, q_{ik} is the quality of the k-th modality for video v_i, and q_k^{thr} is the quality threshold for the k-th modality. In the experimental analysis we provide results also for the simple non-weighted average, with and without likelihood weighting.

Regarding the *Meta-level SVM* and *Meta-level SVM-FSC* fusion methods, the likelihoods output by the uni-modal classifiers are fed into a meta-level classifier. The *Meta-level SVM* fusion uses a standard one-versus-all multi-class SVM. The *Meta-level SVM-FSC* uses a network of binary classifiers similar to the one used in the intermediate fusion but where the inputs are likelihoods instead of features; therefore, the fusion of likelihoods from different modalities is done separately for each class. For both the Meta-SVM and the Meta-SVM-FSC, the SVM parameters are optimized by grid search. In *Meta-SVM+*W_Q and *Meta-SVM-FSC+*W_Q variants the modality qualities are used for weighting the input likelihoods, similarly as for the $weighted\ average + W_Q$.

All the quality thresholds have been optimized by grid search separately for each of the described late fusion methods.

VI. Experimental Analysis

We have extensively tested our proposed approach on real test data. In this section we describe the experiments and discuss the obtained results.

A. Data Collection

As there is no publicly available dataset of mobile user generated video sequences with associated auxiliary sensor data, we collected our own dataset for performing the experimental analysis. We used commercial smartphones which are capable of video and audio recording and which embed the considered auxiliary sensors–magnetometer and accelerometer. In particular, we used four different models of smartphones, each having different characteristics in terms of camera sensor, microphones, and accelerometer and magnetometer sensors. We developed a dedicated software framework that allows for collecting sensor data simultaneously with video recording at sampling rates of 10 and 40 samples per second for the magnetometer and the accelerometer, respectively. Each sample of auxiliary sensor data is temporally aligned to the audio-visual stream by using its timestamp. In order to understand which input videos belong to which sport event we perform temporal alignment of data streams captured by different devices. However, the alignment itself is outside the scope of this paper.



Fig. 7. Sample frames extracted from each sport type C_i considered in our video dataset.

TABLE I
Dataset of Sport Types Used in the Experiments

Sport type	Number of events	Number of users	Number of videos	Video data length (min)
Soccer	6	35	112	1057
Am. football	3	18	62	801
Basketball	3	13	63	492
Tennis	3	12	66	370
Ice-hockey	3	17	86	811
Volleyball	4	17	90	593
TOTALS	**22**	**112**	**479**	**4124**

The data was collected at 22 different public sport events: 6 *soccer* games (class C_0), 3 *American football* games (C_1), 3 *basketball* games (C_2), 3 *tennis* matches (C_3), 3 *ice-hockey* games (C_4) and 4 *volleyball* games (C_5). More details are given in Table I. Sample video frames from our dataset are shown in Fig. 7.

For each event we asked a number of users to record videos using smartphones in which our multimodal data recording software was installed. We did not give any specific instructions on how to perform video recording. On the contrary, users were asked to record the event as they would normally do for their personal use of the videos, within a time-slot of about 1 hour per event. The users were located uniformly in the spectator areas. Their technical skills related to video recording were quite varying, as well as their understanding of the rules for the recorded sport.

B. Methodology for Performance Evaluation

In order to assess the performance of our approach we adopted the leave-one-out (LOO) cross-validation technique. Videos belonging to the same event can be correlated, even those recorded by different users. This is true not only for the

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visual data, but also for the audio and sensor data associated to the videos. Such correlation may result in biasing the obtained classification accuracies. In order to remove such bias, instead of leaving out one video, at each LOO iteration we left out one event, i.e., all videos and associated data from that event.

Another potential source of bias could be that some events of the same sport type were held in the same venue. However there were significant differences in several aspects among these events. For example there were differences in the time of the day and the weather conditions; as a consequence, the lighting conditions and thus the visual features changed considerably. The amount of spectators was always different, especially when considering the area around each camera–therefore the recorded audio scene is different for each event. The users and their spatial location within the venue varied a lot too–thus the auxiliary sensor data, which describes how the cameras are moved, was different for each event. Furthermore, some events of different sport type were held in the same venue, hence making the classification problem even harder–this happened for example for some of the volleyball and basketball games.

The focus of this paper is on classifying an entire event, which consists of all the videos recorded by all the users present at that event; thus our main interest is in maximizing the *event-level accuracy*, referred to as Acc_E and computed as follows: $Acc_E = 100 \times (N_E^c/N_E)$, where N_E^c and N_E are the number of correctly classified events and the total number of events, respectively. However, we provide accuracies also for classifying the single videos of an event, i.e., *video-level accuracies*, which are referred to as $\mathrm{Acc_V}$ and are computed as follows: $\mathrm{Acc_V} = 100 \times (N_V^c/N_V)$, where N_V^c and N_V are the number of correctly classified videos and the total number of videos, respectively. By comparing video- and event- level accuracies it is possible to show how effectively videos from multiple users are combined for classifying events. High $\mathrm{Acc_E}$ and low $\mathrm{Acc_V}$ would mean that the correctly classified videos are well distributed over all events. Instead, low $\mathrm{Acc_E}$ and high $\mathrm{Acc_V}$ would mean that most of the correctly classified videos belong to few events. For a certain $\mathrm{Acc_V}$, the higher $\mathrm{Acc_E}$ is achieved, the more effective the aggregation from video-level to event-level has been.

C. Proposed Features

Fig. 8 shows histograms of average panning extent and compass (i.e., magnetometer) variance values averaged over all videos of each sport type. As can be seen, for some sports the sensor features are very discriminative. For example, high average panning extent characterizes basketball and ice-hockey sports, in which the main action moves "smoothly" from one side of the field to the other, thus triggering panning movements of high angular extent. Instead, in soccer, the players try to reach the goal area advancing by short steps, thus they trigger panning movements of limited extent. In tennis the action moves smoothly but the court is small; consequently, the panning extents are limited too.

Fig. 9 shows STIP features extracted from a segment identified as stable by the sensor analysis and from a non-stable segment. It is clearly visible that features from stable segments are much less noisy because they were detected mainly from the players. Instead, features extracted from non-stable segments were detected also from static structures, due to camera motion. The percentage of stable video content, computed over all

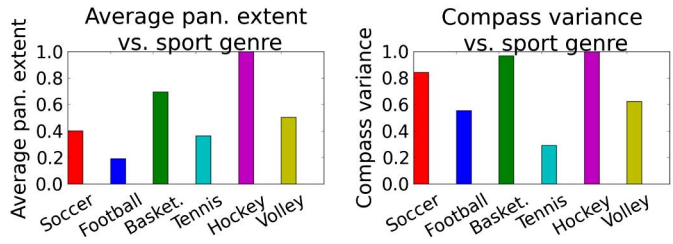


Fig. 8. Average sensor feature values (computed using the auxiliary sensor data associated to all videos in our dataset) for each considered sport type. "Hockey" stands for ice-hockey.

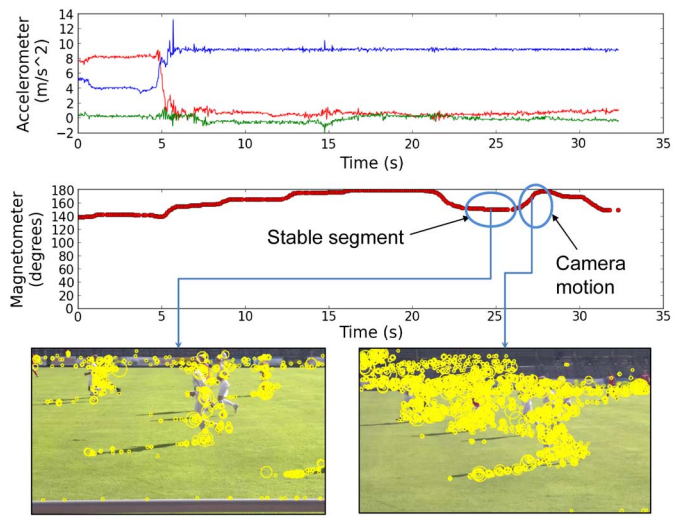


Fig. 9. Spatio-temporal visual features extracted from stable and non-stable video segments, and the associated magnetometer and accelerometer data. When the camera is steady, the extracted interest points are much less noisy (first frame) than when the camera moves (second frame).

videos of all events in our dataset, is only 27.6%; thus, we also avoid performing a huge amount of computation.

Fig. 10 and Fig. 11 provide classification accuracies obtained by using only STIP features, extracted by the proposed sensor-aided approach and by the original approach, respectively, and for all combinations of HOG and HOF vocabulary sizes. Features extracted using the sensor-aided approach achieve higher accuracies at both event-level and video-level–see Fig. 10. This demonstrates the advantage of computing spatio-temporal features from stable video segments only. Furthermore, by way of comparison, we evaluated a BOW setup with a *single vocabulary* for both HOG and HOF descriptors: the best accuracies for each approach are reported in Table II. It is clear that the classification benefits from using separate vocabularies.

D. Performance Evaluation

Firstly, we computed the uni-modal accuracies by considering only one data modality at a time–only sensors, only audio, only spatial visual, only sensor-aided spatio-temporal visual; these accuracies can be found in Table III. Then, we checked whether the errors made by the uni-modal classifiers are correlated. First, for each class $\mathrm{C_j}$, we built one error sequence per modality: $\boldsymbol{e}_j^s, \boldsymbol{e}_j^a, \boldsymbol{e}_j^v, \boldsymbol{e}_j^t$. Each error sequence consists of N_j binary values, where N_j is the number of videos belonging to class $\mathrm{C_j}$, and each element e_{ji} is equal to 1 if the video v_i is

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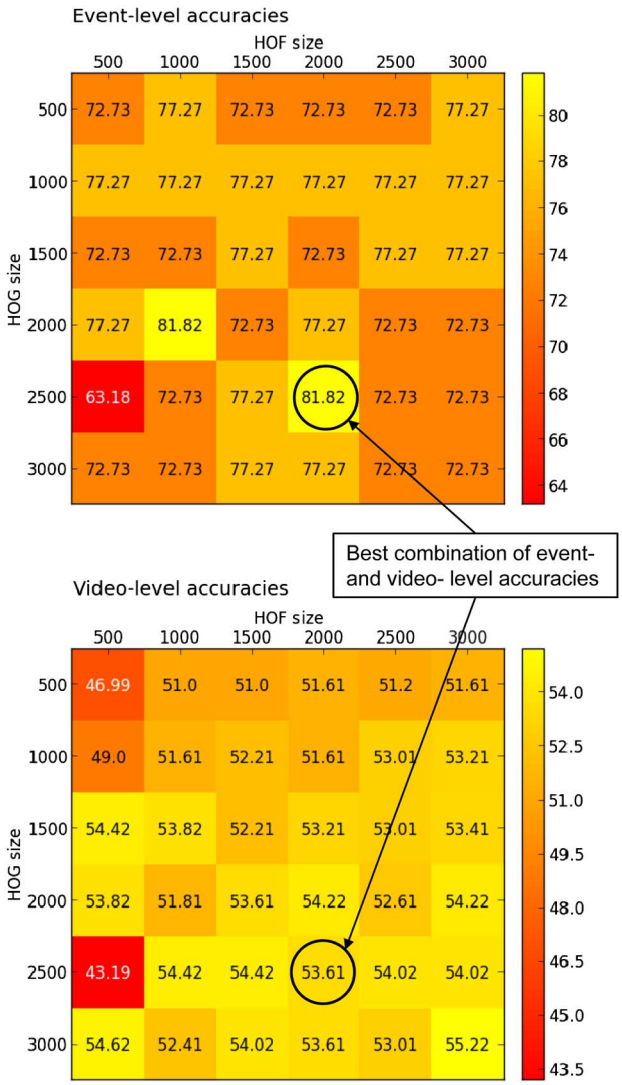


Fig. 10. Event- (top) and video- (bottom) level accuracies obtained by classifying STIP features extracted only from stable segments, for different combinations of sizes for the HOG and HOF BOW vocabularies.

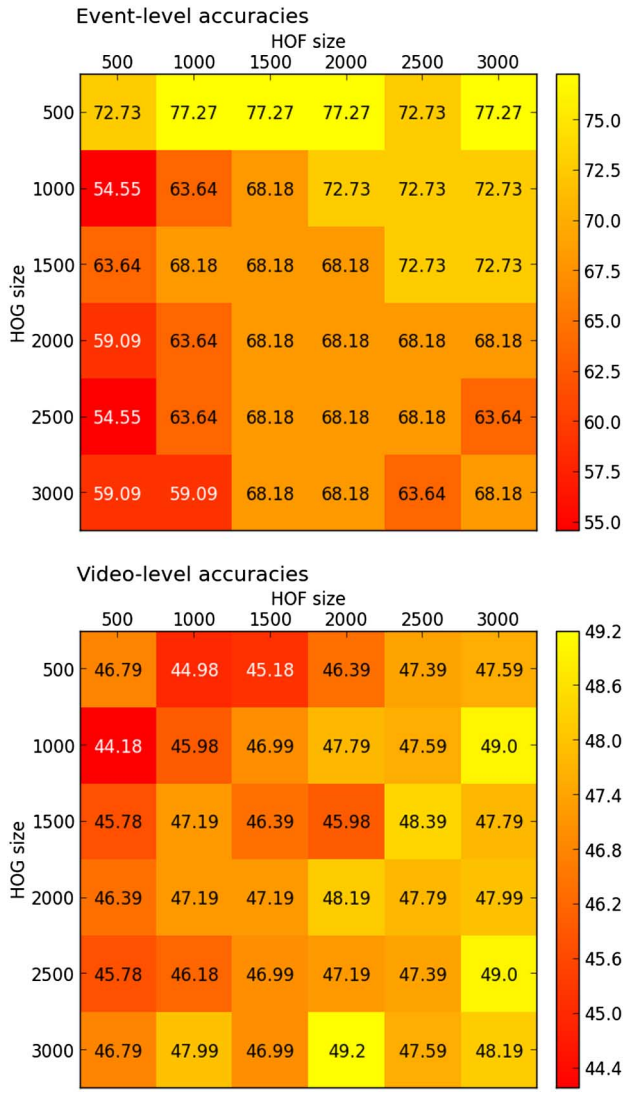


Fig. 11. Event- (top) and video- (bottom) level accuracies obtained by classifying STIP features extracted with the original method, for different combinations of sizes for the HOG and HOF BOW vocabularies.

misclassified, 0 otherwise. The percentages of correlated errors for each class C_j were computed as follows:

$$E_j^{corr} = 100 \times \frac{\sum_{i=1}^{N_v} e_{ji}^{corr}}{\sum_{i=1}^{N_v} e_{ji}^{tot}}$$
$$where \quad e_{ji}^{corr} = e_{ji}^{s} \wedge e_{ji}^{a} \wedge e_{ji}^{v} \wedge e_{ji}^{t}$$
$$and \quad e_{ji}^{tot} = e_{ji}^{s} \vee e_{ji}^{a} \vee e_{ji}^{v} \vee e_{ji}^{t}. \qquad (5)$$

e_{ji}^{corr} indicates the presence of a correlated error among all modalities for video v_i, e_{ji}^{tot} indicates the presence of an error in at least one modality for video v_i, $\wedge$ is the binary AND operator and $\vee$ is the binary inclusive OR operator. The computed percentages for all classes are as follows: $E_0^{corr} = 8.60\%$, $E_1^{corr} = 1.72\%$, $E_2^{corr} = 31.75\%$, $E_3^{corr} = 1.69\%$, $E_4^{corr} = 1.20\%$, $E_5^{corr} = 4.11\%$. As can be seen, apart from class C_2, the errors are weakly correlated. This indicates a potential for improvement of the classification accuracies by fusing the different modalities.

Then, we evaluated the performance of fusing different modalities, by considering all combinations of two modalities (reported in Fig. 12), three and four modalities (Fig. 13), using all the considered fusion strategies, with and without using modality quality levels. The majority voting fusion makes sense only when the number of modalities is greater than two; therefore we used it only for fusing three and four modalities. The abbreviations used in Fig. 12 and Fig. 13 for referring to fusion methods are explained in Table IV. In the figures, the accuracies appearing in bold text indicate that the fusion has succeeded, because they are higher than the highest uni-modal accuracy of the contributing modalities.

TABLE II
COMPARISON BETWEEN THE USE OF A SINGLE AND SEPARATE VOCABULARIES FOR HOG AND HOF

Approach	Single vocabulary		Separate vocabularies	
	Acc_E (%)	Acc_V (%)	Acc_E (%)	Acc_V (%)
STIP from stable segments	77.27	47.29	81.82	53.61
STIP from any segment	68.18	46.79	77.27	47.59

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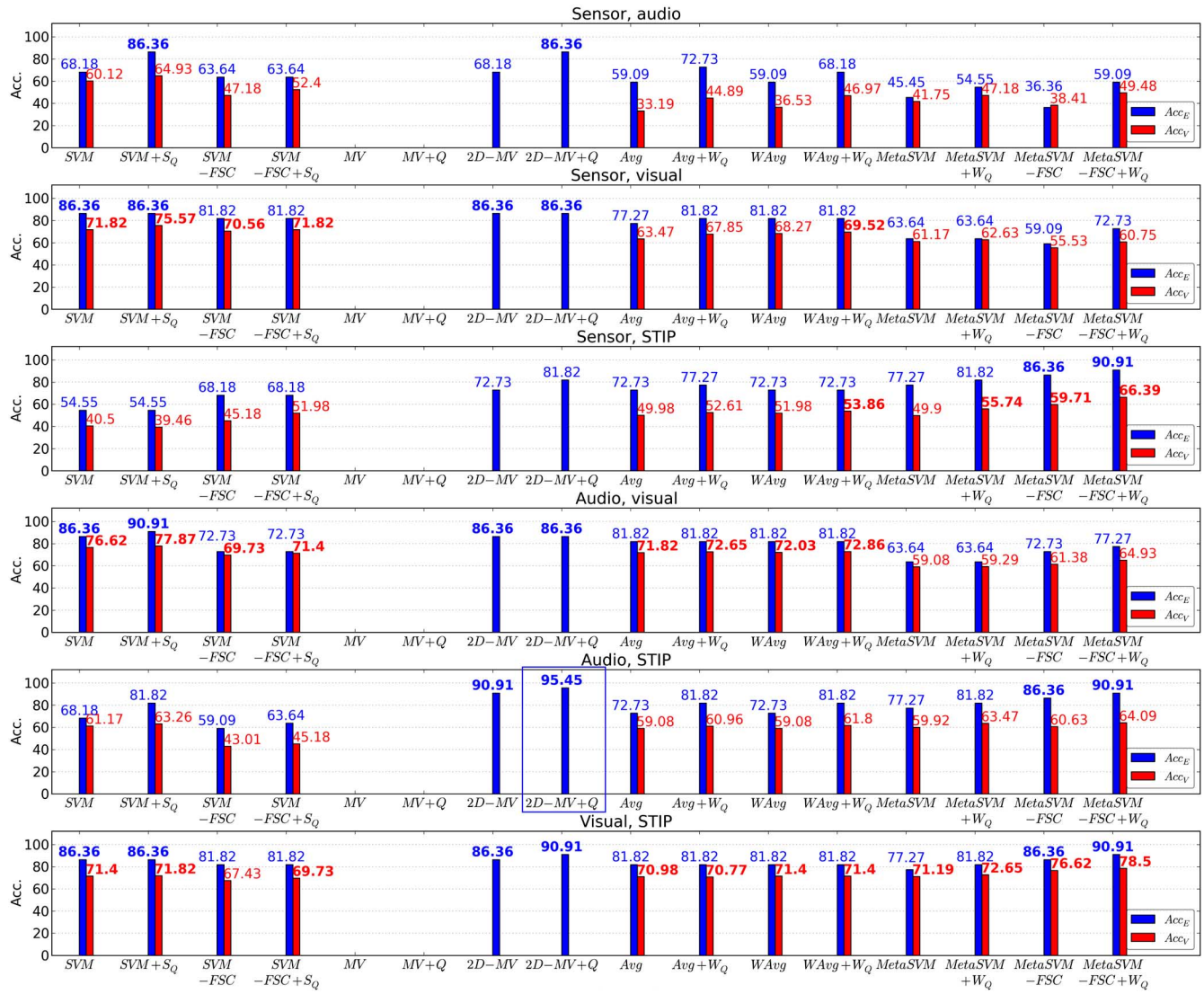


Fig. 12. Classification accuracies for all the considered fusion methods (on the x-axis) and for combinations of two modalities (each plot represents a different combination of modalities). Accuracies in bold font are those which improve over the uni-modal accuracies of the contributing modalities. The blue rectangle highlights the best event-level accuracy.

E. Discussion and Observations

The most important findings from our experimental evaluation can be summarized as follows. (i) The normal and the 2D version of the majority voting fusion are often among the most successful fusion strategies, despite their simplicity. (ii) Late fusion provides the best event-level accuracies. (iii) Late fusion handles features of very different dimensionality in a better way than early fusion. (iv) In rule-based late fusion methods, considering the likelihoods of all classes often brings more confusion than considering only the most likely class. (v) In the fusion step, standard SVM performs better in early fusion than in late fusion. Instead, SVM-FSC is more successful in late fusion than when applied directly on features. (vi) The proposed quality-based fusion adaptation improves the accuracies for most tested cases. (vii) The highest event-level accuracy is achieved only when using also the STIP modality. (viii) One example of good trade-off between accuracy and complexity is represented by combining sensor and audio using 2D majority voting fusion. (ix) The addition of the sensor modality improves the accuracies in many cases, despite its low computational footprint. (x) As it will be shown in the next subsections, the proposed fusion is able to increase classification accuracies even when it considers only videos from the short sides of the sport venue or only videos recorded by one user.

In the following we will elaborate more on the achieved results and related findings. The best event-level accuracy, which is $\mathrm{Acc_E} = 95.45\%$ is obtained by the majority voting + Q and by the 2D majority voting + Q fusion methods applied on different combinations of modalities. These cases are highlighted in blue in Fig. 12 and Fig. 13.

The best video-level accuracy, $\mathrm{Acc_V} = 79.75\%$, is achieved by the $\mathrm{SVM} + \mathrm{S_Q}$ applied on all modalities–this case is highlighted in red in Fig. 13.

A good pair of event- and video- level accuracies is provided by the majority voting + Q fusion applied on all modalities, which achieves $\mathrm{Acc_E} = 95.45\%$ and $\mathrm{Acc_V} = 75.78\%$, and is highlighted in green in Fig. 13. For this case, a confusion matrix is provided in Table V. The main confusion at both

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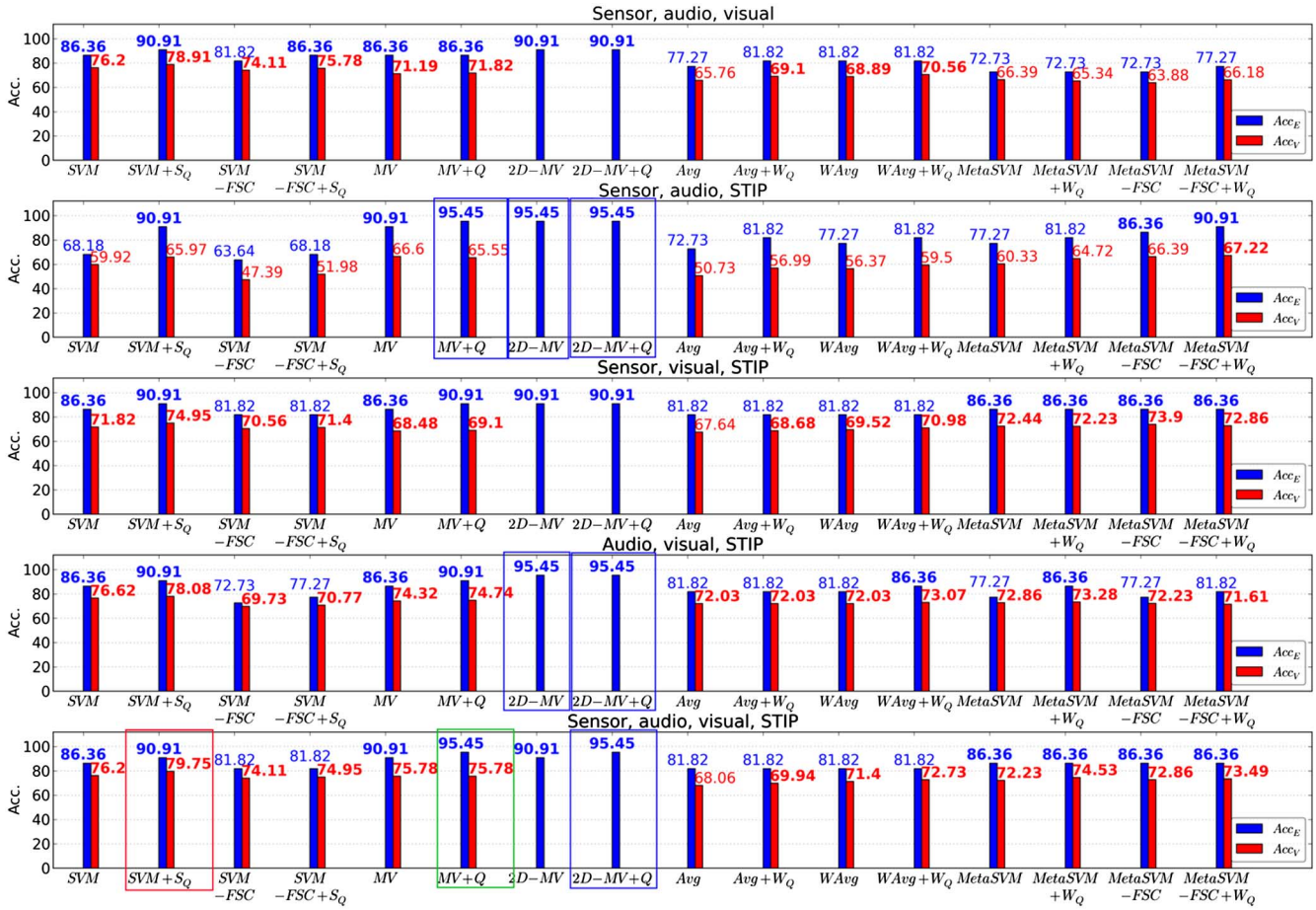


Fig. 13. Classification accuracies for all the considered fusion methods (on the x-axis) and for combinations of three modalities (each plot represents a different combination of modalities). Accuracies in bold font are those which improve over the uni-modal accuracies of the contributing modalities. Blue and red rectangles highlight the best event- and video- level accuracies, respectively, whereas the green rectangle highlights the best pair of event- and video- level accuracies.

TABLE III
UNI-MODAL ACCURACIES

Modality	Acc_E (%)	Acc_V (%)
Sensor	54.55	34.31
Audio	72.73	67.07
Spatial visual	81.82	68.45
Spatio-temporal visual	81.82	53.61

TABLE IV
ABBREVIATIONS USED IN FIG. 12 AND FIG. 13 FOR REFERRING TO FUSION METHODS

Fusion method	Abbreviation
Early fusion	
SVM	*SVM*
SVM using qualities for weighting features	*SVM+S_Q*
Intermediate fusion	
SVM fusing separately for each class	*SVM-FSC*
SVM fusing separately for each class and using qualities for weighting features	*SVM-FSC+S_Q*
Late fusion	
Majority voting	*MV*
Majority voting using qualities to select modalities	*MV+Q*
2D majority voting	*2D-MV*
2D majority voting using qualities to select modalities	*2D-MV+Q*
Average	*Avg*
Average using qualities for weighting probabilities	*Avg+W_Q*
Weighted average	*WAvg*
Weighted average using qualities for weighting probabilities	*WAvg+W_Q*
Meta-level SVM	*MetaSVM*
Meta-level SVM using qualities for weighting probabilities	*MetaSVM+W_Q*
Meta-level SVM fusing separately for each class	*MetaSVM-FSC*
Meta-level SVM fusing separately for each class and using qualities for weighting probabilities	*MetaSVM-FSC+W_Q*

event- and video- level is on basketball (C_2), which is confused as volleyball (C_5). This result was expected for two reasons: first, the venues for these two sports have similar visual appearance; second, the correlation of uni-modal errors for class C_2 was high (31.75%), hence the different modalities do not always provide sufficiently complementary information for this class. At the video level there is significant confusion between American football (C_1) and soccer (C_0)–probably due to the similar colors of the playing fields–, and between tennis (C_3) and American football (C_1).

For each combination of modalities, Table VI summarizes the fusion strategies which provide the best Acc_E, the best Acc_V, or the pair (Acc_E, Acc_V) representing the best trade-off. The best trade-off is represented by the fusion method which achieves the highest Acc_E among all the fusion methods, as we are more concerned about classifying the whole event. In case multiple methods yield the highest Acc_E, the best trade-off is the method

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TABLE V
CONFUSION MATRIX FOR CLASSIFYING THE SPORT TYPE OF EVENTS (AND, IN PARENTHESIS, OF VIDEOS) USING THE MAJORITY VOTING +Q FUSION ON ALL MODALITIES. GT STANDS FOR GROUND TRUTH LABELS AND PR. STANDS FOR PREDICTIONS. CLASSIFICATION RESULTS ARE EXPRESSED AS PERCENTAGES

Pr. / GT	C_0	C_1	C_2	C_3	C_4	C_5
C_0	**100 (80.36)**	0 (8.04)	0 (4.46)	0 (2.68)	0 (2.68)	0 (1.79)
C_1	0 (12.90)	**100 (77.42)**	0 (0)	0 (6.45)	0 (1.61)	0 (1.61)
C_2	0 (1.59)	0 (3.17)	**66.67 (52.38)**	0 (0)	0 (3.17)	33.33 (39.68)
C_3	0 (1.51)	0 (15.15)	0 (0)	**100.0 (80.30)**	0 (1.51)	0 (1.51)
C_4	0 (11.63)	0 (0)	0 (5.81)	0 (0)	**100 (79.07)**	0 (3.49)
C_5	0 (0)	0 (1.11)	0 (7.78)	0 (0)	0 (1.22)	**100 (78.89)**

TABLE VI
BEST FUSION METHODS FOR EACH COMBINATION OF MODALITIES, ACCORDING TO $\mathrm{Acc_E}$, $\mathrm{Acc_V}$, AND THE PAIR ($\mathrm{Acc_E}$, $\mathrm{Acc_V}$). S, A, V, T STAND FOR SENSOR, AUDIO, SPATIAL VISUAL AND SPATIO-TEMPORAL VISUAL

Modality combination	Best $\mathrm{Acc_E}$	Best $\mathrm{Acc_V}$	Best ($\mathrm{Acc_E}$, $\mathrm{Acc_V}$)
S, A	SVM+S_Q, 2D-MV+Q	SVM+S_Q	SVM+S_Q
S, V	SVM, SVM+S_Q, 2D-MV, 2D-MV+Q	SVM+S_Q	SVM+S_Q
S, T	MetaSVM-FSC+W_Q	MetaSVM-FSC+W_Q	MetaSVM-FSC+W_Q
A, V	SVM+S_Q	SVM+S_Q	SVM+S_Q
A, T	2D-MV+Q	MetaSVM-FSC+W_Q	MetaSVM-FSC+W_Q
V, T	2D-MV+Q, MetaSVM-FSC+W_Q	MetaSVM-FSC+W_Q	MetaSVM-FSC+W_Q
S, A, V	SVM+S_Q, 2D-MV, 2D-MV+Q	SVM+S_Q	SVM+S_Q
S, A, T	MV+Q, 2D-MV, 2D-MV+Q	MetaSVM-FSC+W_Q	MV+Q
S, V, T	SVM+S_Q, MV+Q, 2D-MV, 2D-MV+Q	SVM+S_Q	SVM+S_Q
A, V, T	2D-MV, 2D-MV+Q	SVM+S_Q	SVM+S_Q
S, A, V, T	MV+Q, 2D-MV+Q	SVM+S_Q	MV+Q

achieving the (relative) highest $\mathrm{Acc_V}$ among those with highest $\mathrm{Acc_E}$.

1) Influence of the Viewing Angle: Data captured by users who are in distinct locations of the spectator areas are likely to have different characteristics due to the different viewing perspective. This is especially true when considering the perpendicular perspectives, e.g., the long and the short sides of a soccer field. For example, camera motion would be different on the short side of a soccer field compared to the long side–see Fig. 14 which shows the average panning extent feature separately for videos recorded from the short and from the long sides.

In our approach we combine the information from all videos captured from all sides in order to obtain the event-level prediction. However, in practice it may happen that all the users have recorded from similar angles, e.g., only from the short side.

In order to show the influence of the viewing angle on the classification accuracy, we computed the accuracies separately for each of the two perpendicular viewing angles. We did this for the fusion case which achieves the best pair of event- and

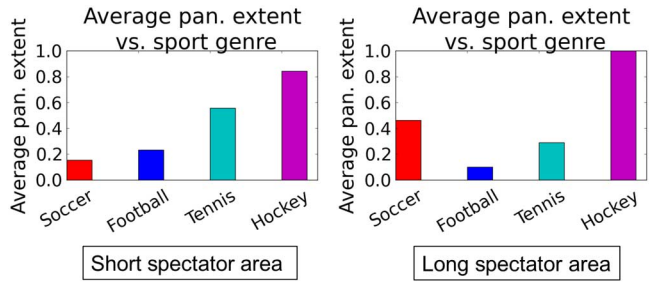


Fig. 14. Average panning extent feature computed separately for short and long spectator sides. Only the events with both short and long sides were considered.

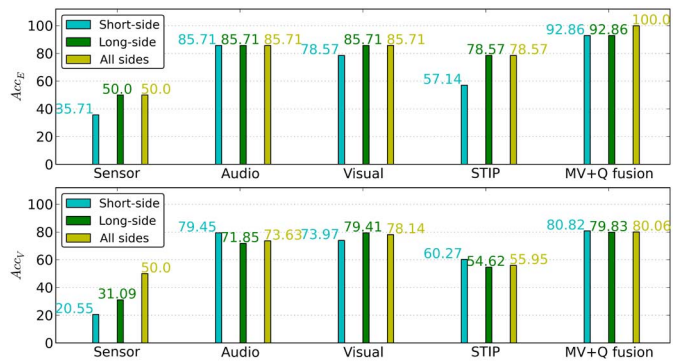


Fig. 15. Accuracies computed separately for each side of the sport venues.

video- level accuracies, i.e., fusing all the modalities using majority voting + Q. In this experiment we have considered only those events having both short and long spectator areas, and left out those other events with only long side; thus, for the sake of comparison, we recomputed the accuracies also for the normal case of using the videos from all the sides, by considering only this subset of events. Fig. 15 shows all these accuracies, for each contributing modality alone and for their fusion. Short-side videos yield lower event-level accuracies than long-side videos in most uni-modal cases. Thus, if multimodal fusion is not performed, features extracted from short-side videos are not as discriminative as for long-side data. However, in the non-optimal case in which only videos from the short side are available for analysis, the fused accuracy is still high (92.86%), thanks to both the fusion of modalities and the aggregation of video predictions. The accuracy obtained by analyzing all-sides videos is not negatively affected by the short-side videos. On the contrary, when fusing all modalities captured from all sides, the event accuracy improves over the single-side event accuracies.

2) Fusion of Videos from a Single User: We wanted to analyze also how well the sport type of the event is classified if only the videos from one user were available. Thus, we computed the per-user accuracy $\mathrm{Acc_{User}}$, which is the percentage of users for which the event recorded by each user was classified correctly. In Fig. 16, $\mathrm{Acc_V}$ represents how well the sport type is classified if we analyzed only one video, whereas $\mathrm{Acc_{User}}$ represents how well the sport type is classified if we analyzed the videos recorded only by a certain user in an event. The aggregation of video predictions is beneficial also when applied to videos from only one user; this is true for both the case of considering only one modality and the case of fusing all modalities. By performing both multimodal fusion and aggregation of video

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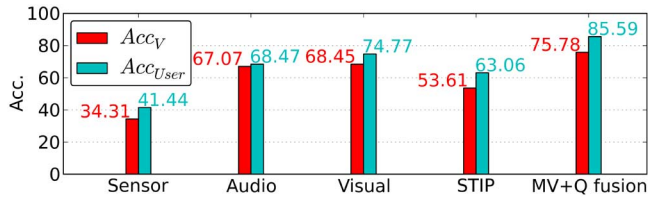


Fig. 16. Comparison between per-video accuracies and per-user accuracies.

predictions, we correctly classify the sport type for 85.59% of users.

3) Comparison of Early, Intermediate and Late Fusion Approaches: Since early fusion is applied directly on the features, it is sensitive to their dimensionality. STIP features have very high dimensionality compared especially to sensor and audio modalities, thus influencing the fusion process significantly. Also, the STIP modality yields relatively low accuracy. As a consequence, if STIP features are fused with sensor and/or audio features only, the performance is very low, because STIP is the predominant modality. Instead, if STIP is used with spatial visual features then early fusion performs well. If STIP features are not used at all, early fusion is the best performing fusion approach. Furthermore, early fusion with feature weighting achieves the highest video-level accuracies.

Regarding the intermediate fusion, as can be seen in Fig. 13, $\mathrm{SVM-FSC+S_Q}$ improves the video accuracy up to 75.78%, but it improves the event accuracy only in one case.

Late fusion is the best fusion approach in aggregating video-level fused predictions to event-level. In fact, majority voting and 2D majority voting achieve an event-level accuracy of 95.45%, even when fusing only two modalities, i.e., audio and STIP. In particular, applying majority voting + Q on sensor, audio and STIP modalities allows for an aggregation which increases $\mathrm{Acc_V} = 65.55\%$ to $\mathrm{Acc_E} = 95.45\%$, thus bringing an improvement of 29.9 percentage points. These late fusion methods achieve high event-level accuracies because they effectively exploit complementarities among modalities for a sufficient number of videos and for most events.

Using SVM in fusion achieves different results depending on what they are applied to, i.e., features or classifiers' outputs. The standard one-versus-all multi-class SVM performs well in early fusion, whereas it performs worse in late fusion. Therefore, standard SVM is able to exploit the complementarities between modalities in a better way when working directly on the features. Instead, SVM-FSC has the opposite behavior–it performs better when applied on the likelihoods than when applied on the features. The main peculiarity in the FSC variant of SVM is that it fuses different modalities (either features or likelihoods) separately for each class. Thus, SVM-FSC performs better in late fusion probably because the likelihoods of different modalities, which already encode separate information for each class, are more complementary than raw features. One example of successful SVM-FSC in late fusion is the Meta-SVM-FSC + $\mathrm{W_Q}$ applied on spatial visual and STIP modalities, which achieves $\mathrm{Acc_E} = 90.91\%$ and $\mathrm{Acc_V} = 78.50\%$.

4) Comparison of Classifier-Based and Rule-Based Fusion: The fusion methods which improve the uni-modal accuracies in most of the modality combinations are the 2D majority voting + Q, the majority voting, the SVM + $\mathrm{S_Q}$, and the Meta-SVM-FSC + $\mathrm{W_Q}$. However, SVM + $\mathrm{S_Q}$ and Meta-SVM-FSC + $\mathrm{W_Q}$ are supervised fusion techniques, whereas majority voting is unsupervised. Therefore, if new sport types need to be added to the classification task, supervised fusion methods need to be retrained, whereas the majority voting fusion does not require any re-training. Furthermore, the computational cost of SVM-based fusion is likely to be higher than for majority voting.

The other rule-based fusion methods, i.e., average and weighted average, often achieve accuracies which are even lower than the uni-modal accuracies. This means that summing up likelihoods from different uni-modal accuracies results in obtaining the maximum score for an incorrect class label. This may be due to the uncertainty of the uni-modal classifiers, for which high likelihood values are output for multiple classes instead of only for the most likely class. Instead, in majority voting fusion, for each video only the class with highest likelihood is considered, and the uncertainty represented by the likelihoods of the other classes is completely ignored, thus avoiding their influence in the fusion process. Therefore, considering the likelihoods for all classes, as it is done in average and weighted average fusion methods, often brings more confusion instead of useful information.

5) Use of Modality Quality Levels: The results in Fig. 12 and Fig. 13 show that adapting the fusion process according to the modality quality levels improves the classification accuracies for most tested cases. In fact, either the $\mathrm{Acc_E}$ or the $\mathrm{Acc_V}$ (or both) are improved in most modality combinations by $\mathrm{SVM+S_Q}$, by majority-voting+Q, by weighted-average+$\mathrm{W_Q}$, by Meta-SVM + $\mathrm{W_Q}$, and by Meta-SVM-FSC + $\mathrm{W_Q}$. The 2D-majority-voting+Q improves the event accuracy in 5 out of 11 combinations.

6) Discussion on the Considered Modalities: We measured the processing time required for extracting features, separately for each modality. In the classification phase, the feature extraction represents the most time-consuming step, thus the measured processing times give indications also about the total classification time. We considered one video of about 10 minutes for each sport class. The processing times averaged over all sport classes are as follows: 2100 seconds for spatio-temporal visual analysis; 150 seconds for the spatial visual analysis; 11 seconds for the audio analysis; 0.51 seconds for the sensor analysis. We performed the tests on a machine equipped with 92 GB of RAM and an 8-core 2.53 GHz processor. The low complexity of the sensor modality is partly due to the light-weight nature of its data. In fact, one second of auxiliary sensor data consists of about 50 samples, whereas one second of HD video content contains about 23 million pixels.

Two good trade-offs between event accuracy and complexity are represented by applying 2D majority voting + Q for fusing either sensor and audio modalities, or sensor, audio, and spatial visual modalities. A good trade-off between video accuracy and complexity is the fusion of sensor, audio and spatial visual modalities using SVM + $\mathrm{S_Q}$.

Event accuracy of 95.45% is achieved only by combinations which include the STIP modality, thus confirming that spatio-temporal information plays an important role.

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The addition of the sensor modality improves either the video- or event-level accuracies in many cases, which is a big benefit especially when considering the negligible computational overhead introduced by this modality.

VII. Conclusions

This paper proposes a framework for classifying the sport type in user generated content by analyzing multiple videos and multiple modalities. The use of multiple videos provides some level of redundancy, which is helpful for assigning adequate (low) importance to "outlier" videos. Furthermore, when the videos are taken by different users, they describe the recorded scene from multiple viewpoints, thus providing more complete information than just one video. The use of multiple modalities allows for describing the scene in terms of different sensory information. In addition to audio and spatial-visual modalities, we propose to analyze magnetometer and accelerometer data for extracting spatio-temporal visual features that are less affected by camera motion and for extracting sensor-based camera motion features which have a low computational footprint.

For the fusion process, our work considers early, intermediate and late fusion approaches. In particular, we compared standard multi-class SVM with a modified version of multi-class SVM which fuses the modalities separately for each class. Also, we presented a new rule-based late fusion technique, namely 2D majority voting, which proved to achieve the highest event-level accuracy in several cases.

The contribution of sensor, audio, spatial visual and spatio-temporal visual modalities in the fusion is dynamically adapted by considering their quality estimates given the input data. We also proposed to use cross-modal qualities, i.e., the quality of auxiliary sensor data is analyzed for adapting the contribution of the spatio-temporal visual modality. The proposed fusion adaptation is performed by using the modality qualities either for feature weighting, for probability weighting, or for directly selecting the modalities.

The experimental evaluation provided extensive results on real test data by using all the considered fusion methods and modality combinations. We showed that the fusion of modalities and the use of modality quality levels improve the classification accuracy in most cases, achieving 95.45% of event-level accuracy and 79.75% of video-level accuracy in the best cases. The spatio-temporal visual information proved to be fundamental for achieving the highest accuracies. Finally, our system is still able to achieve high classification accuracies even if it analyzes videos recorded only from the short or long side of the sport venue, or only by one user.

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